

Flag 1 – Install Flutter SDK

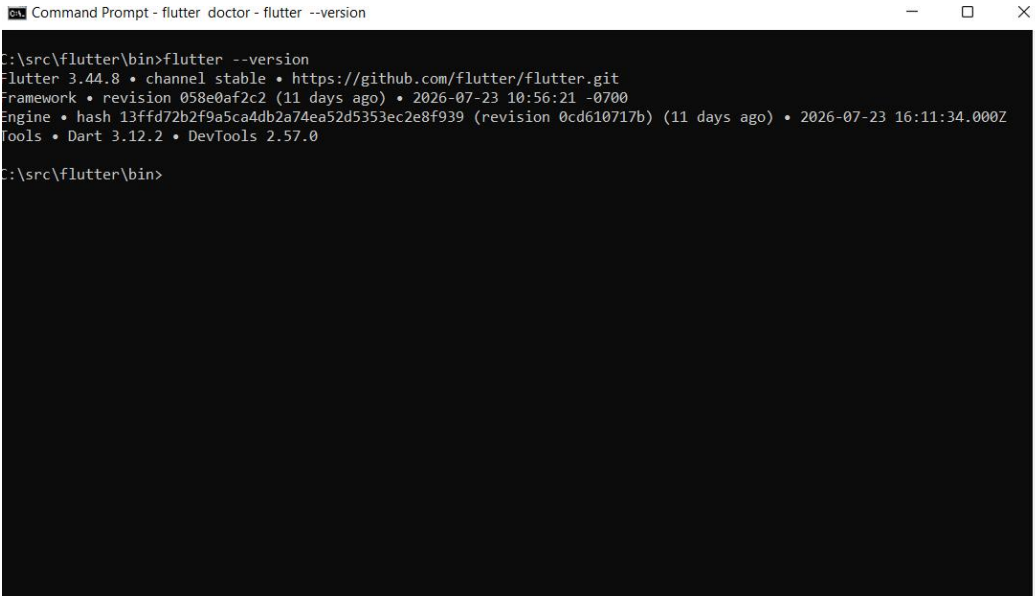
Objective: Install the Flutter SDK.

Verify:

`flutter --version`

Screenshot:

- The results of the command



```
Command Prompt - flutter doctor - flutter --version

C:\src\flutter\bin>flutter --version
Flutter 3.44.8 • channel stable • https://github.com/flutter/flutter.git
Framework • revision 058e0af2c2 (11 days ago) • 2026-07-23 10:56:21 -0700
Engine • hash 13ffd72b2f9a5ca4db2a74ea52d5353ec2e8f939 (revision 0cd610717b) (11 days ago) • 2026-07-23 16:11:34.000Z
Tools • Dart 3.12.2 • DevTools 2.57.0

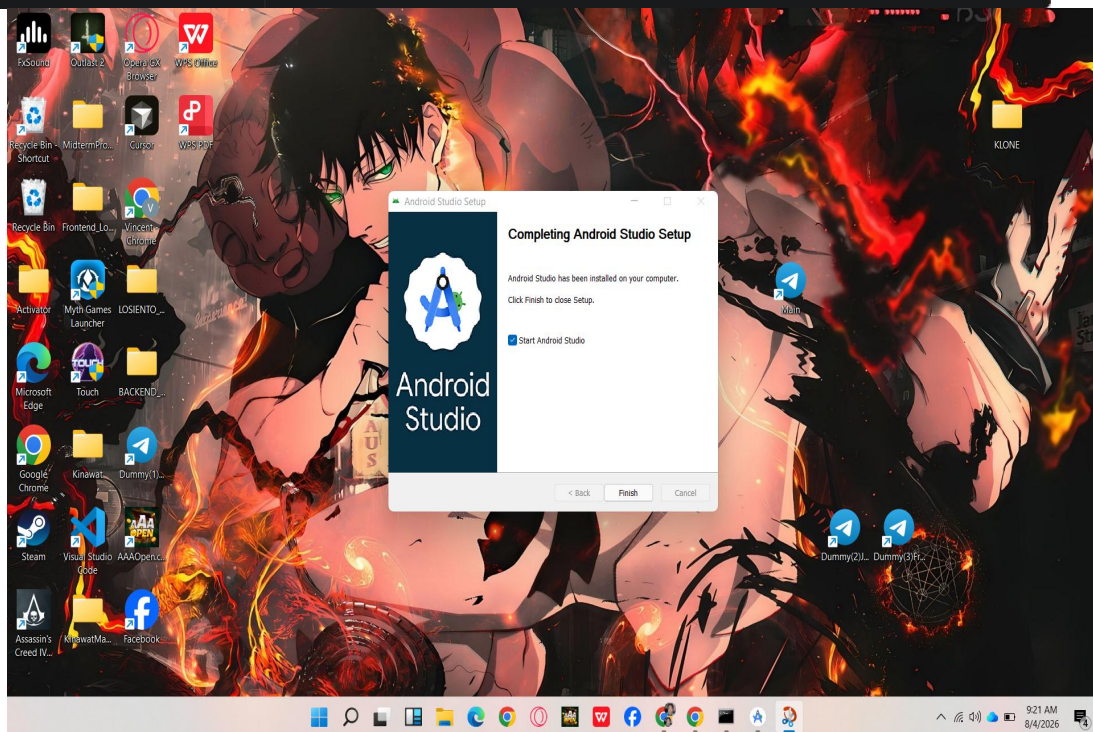
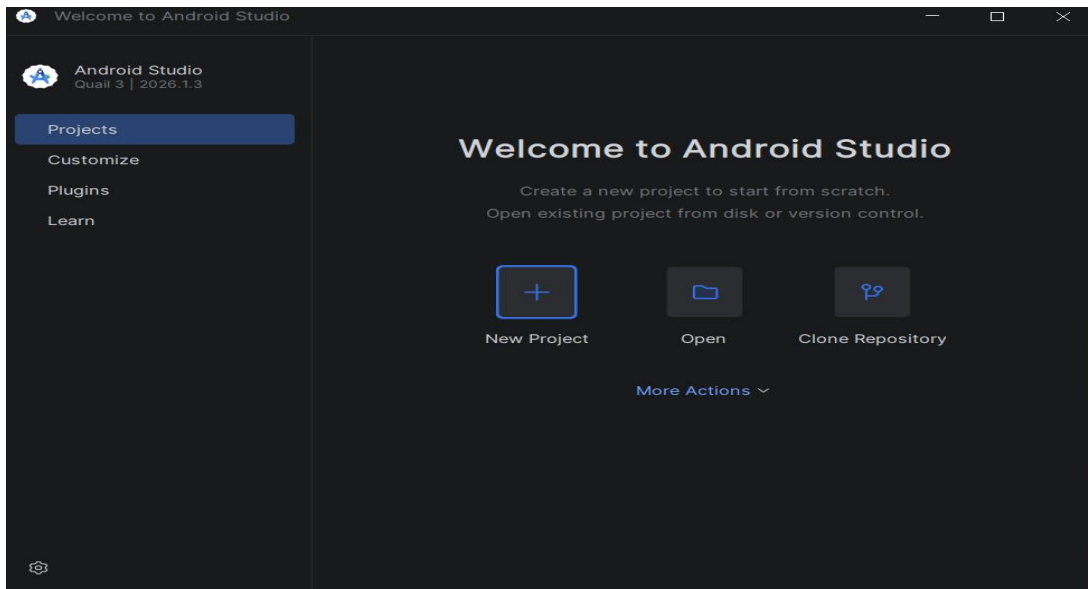
C:\src\flutter\bin>
```

Flag 2 – Install Android Studio

Objective: Install Android Studio.

Screenshot:

- Android Studio installed
- SDK Manager opened (or Welcome screen)

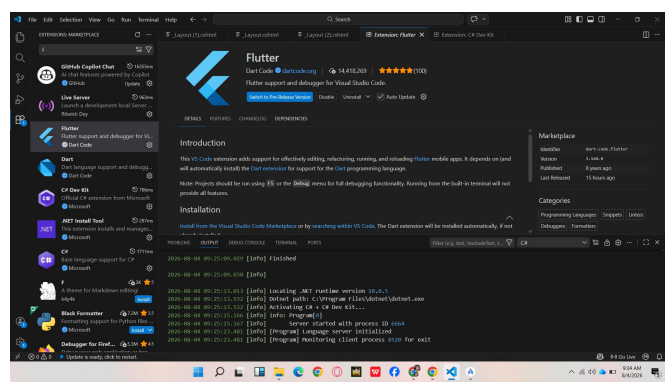
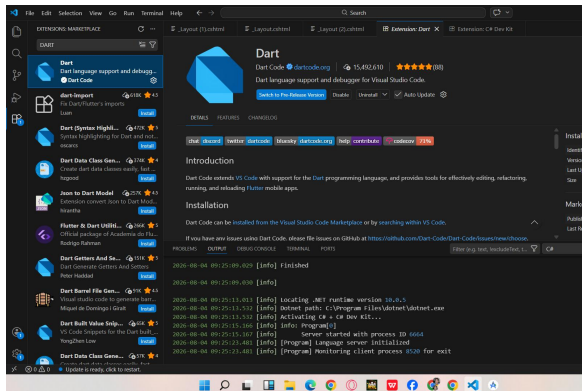


Flag 3 – Install VS Code & Flutter Extension

Objective: Install VS Code and the Flutter (which also installs Dart) extension.

Screenshot:

- Flutter extension installed
- Dart extension installed (or shown as installed)



Flag 4 – Run Flutter Doctor

Objective: Check whether everything is configured correctly.

Run:

flutter doctor

Screenshot:

- flutter doctor results

```
C:\Users\Winja Turtle>flutter doctor
'flutter' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Winja Turtle>C:\src\flutter\bin\flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[!] Flutter (Channel stable, 3.44.8, on Microsoft Windows [Version 10.0.21996.1], locale en-US)
    ! The flutter binary is not on your path. Consider adding C:\src\flutter\bin to your path.
    ! The dart binary is not on your path. Consider adding C:\src\flutter\bin to your path.
[✓] Windows Version (11 Home Single Language 64-bit, Dev, 20H9)
[X] Android toolchain - develop for Android devices
    X cmdline-tools component is missing
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio#command-line-tools-only and make sure
      to set the ANDROID_HOME environment variable.
      See https://developer.android.com/studio/command-line for more details.
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps (Visual Studio Community 2026 18.4.1)
    X Visual Studio is missing necessary components. Please re-run the Visual Studio installer for the "Desktop
      development with C++" workload, and include these components:
        MSVC v142 - VS 2019 C++ x64/x86 build tools
          - If there are multiple build tool versions available, install the latest
        C++ Make tools for Windows
        Windows 10 SDK
[✓] Connected device (3 available)
[✓] Network resources

Doctor found issues in 3 categories.

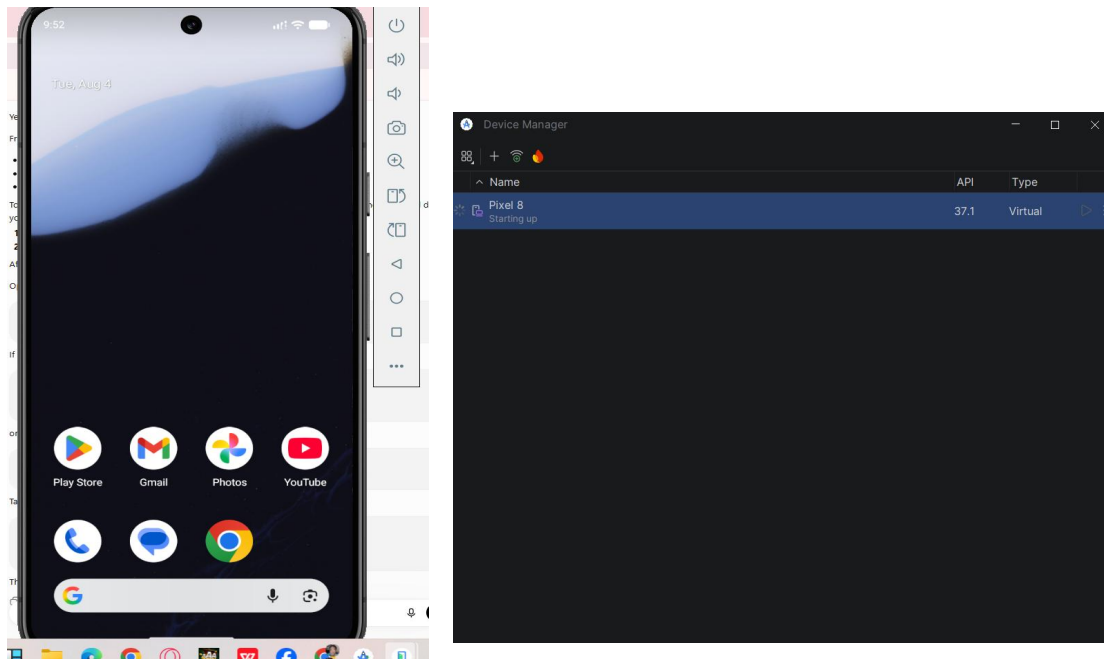
C:\Users\Winja Turtle>
```

Flag 5 – Build Your Virtual Device

Objective: Create your first Android Emulator (AVD).

Screenshot:

- Device Manager
- Your newly created virtual device
- Emulator successfully booted



Flag 6 – Forge Your Project

Objective: Create your first Flutter project.

Run:

```
flutter create my_first_flutter_app
```

Screenshot:

- Successful creation of project

```
C:\Windows\system32\cmd.exe - flutter create C:\Users\Ninja Turtle\Documents\my_first_flutter_app - flutter create "C:\Users\Ninja Turtle\Documents\my_first_flutter_app"

C:\Users\Ninja Turtle>cd C:\src\flutter\bin

C:\src\flutter\bin>flutter create C:\Users\Ninja Turtle\Documents\my_first_flutter_app
Multiple output directories specified.

C:\src\flutter\bin>flutter create "C:\Users\Ninja Turtle\Documents\my_first_flutter_app"
Creating project ..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app...
Resolving dependencies in `..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app`... (1.4s)
Downloading packages...
Got dependencies in `..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app`.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd ..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app
$ flutter run

Your application code is in ..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app\lib\main.dart.

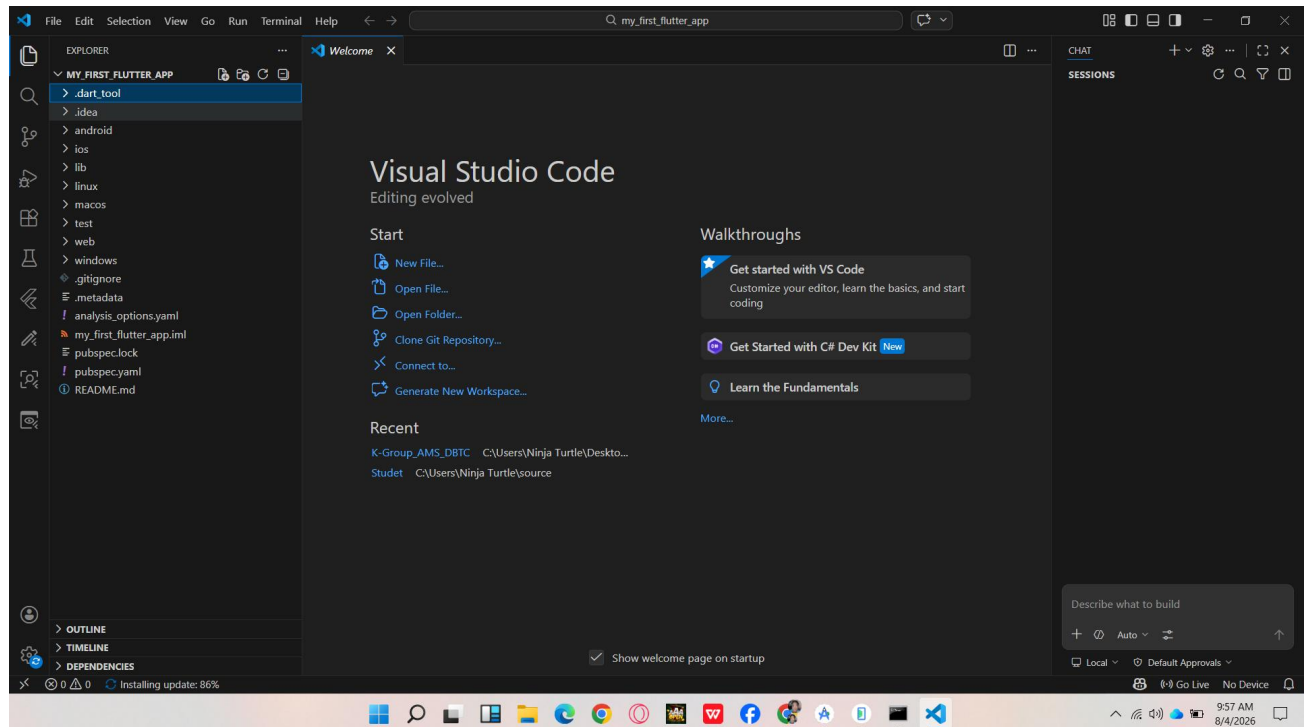
C:\src\flutter\bin>
```

Flag 7 – Unlock the Workspace

Objective: Open the project in VS Code.

Screenshot:

- Your project opened in VS Code



Flag 8 – Launch

Objective: Run your Flutter application.

Run:

flutter run

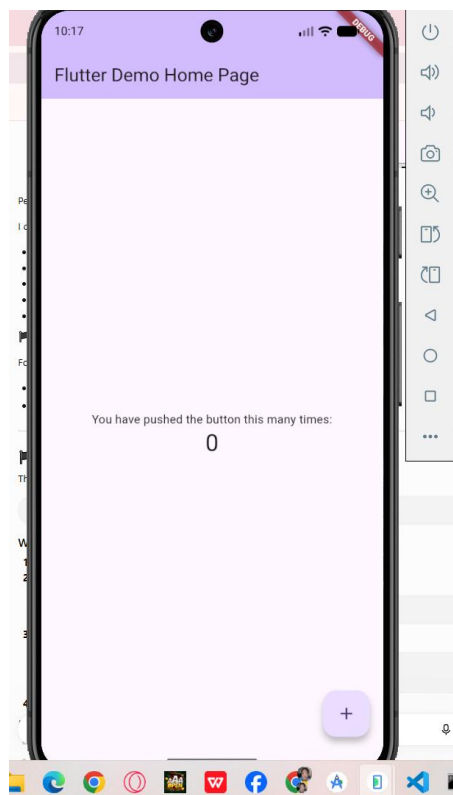
Screenshot:

- Terminal
- Emulator running
- Default Flutter app

```
C:\Windows\System32\cmd.exe - flutter --version C:\src\flutter\bin\flutter run
Launching lib/main.dart on sdk gphone16k x86_64 in debug mode...
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by net.rubygrapefruit.platform.internal.NativeLibraryLoader in an unnamed module (file:/C:/Users/Ninja Turtle/AppData/Local/Android/Sdk\platform-0.22-milestone-28.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

Checking the license for package NDK (Side by side) 28.2.13676358 in C:\Users\Ninja Turtle\AppData\Local\Android\Sdk\licenses
License for package NDK (Side by side) 28.2.13676358 accepted.
Preparing "Install NDK (Side by side) 28.2.13676358 v.28.2.13676358".
"Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" ready.
Installing NDK (Side by side) 28.2.13676358 in C:\Users\Ninja Turtle\AppData\Local\Android\Sdk\ndk\28.2.13676358
"Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" complete.
"Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" finished.
Checking the license for package Android SDK Platform 36 in C:\Users\Ninja Turtle\AppData\Local\Android\Sdk\licenses
License for package Android SDK Platform 36 accepted.
Preparing "Install Android SDK Platform 36 (revision 2)".
"Install Android SDK Platform 36 (revision 2)" ready.
Installing Android SDK Platform 36 in C:\Users\Ninja Turtle\AppData\Local\Android\Sdk\platforms\android-36
"Install Android SDK Platform 36 (revision 2)" complete.
"Install Android SDK Platform 36 (revision 2)" finished.
=====
Checking the license for package CMake 3.22.1 in C:\Users\Ninja Turtle\AppData\Local\Android\Sdk\licenses
License for package CMake 3.22.1 accepted.
Preparing "Install CMake 3.22.1 v.3.22.1".
"Install CMake 3.22.1 v.3.22.1" ready.
Installing CMake 3.22.1 in C:\Users\Ninja Turtle\AppData\Local\Android\Sdk\cmake\3.22.1
"Install CMake 3.22.1 v.3.22.1" complete.
"Install CMake 3.22.1 v.3.22.1" finished.
Running Gradle task 'assembleDebug'... 678.9s
✓ Built build/app/outputs/flutter-apk/app-debug.apk
Installing build/app/outputs/flutter-apk/app-debug.apk... 727ms
D/FlutterActivityAndFragmentDelegate( 8240): If you are attempting to set --enable-dart-profiling via Intent extras to launch a Flutter component outside of using the Flutter CLI, note that support for setting engine flags on Android via Intent will soon be dropped; see https://github.com/flutter/flutter/issues/108686 for more information on this breaking change. To migrate, set --enable-dart-profiling or any other flags specified via Intent extras on the command line instead or see https://github.com/flutter/flutter/blob/main/docs/engine/Flutter-Android-Engine-Flags.md for alternative methods.
D/FlutterJNI( 8240): Beginning load of Flutter...
D/FlutterJNI( 8240): Flutter (null) was loaded normally!
I/Flutter ( 8240): [IMPORTANT:Flutter/shell/platform/android/android_context_gl_implementation.cc(104)] Using the Impeller rendering backend (OpenGL).
D/FlutterRenderer( 8240): Width is zero, 0.0
Syncing files to device sdk gphone16k x86_64... 93ms

Flutter run key commands.
  r Hot reload.
  R Hot restart.
  h List all available interactive commands.
  d Detach (terminate "flutter run" but leave application running).
  c Clear the screen
  q Quit (terminate the application on the device).
```

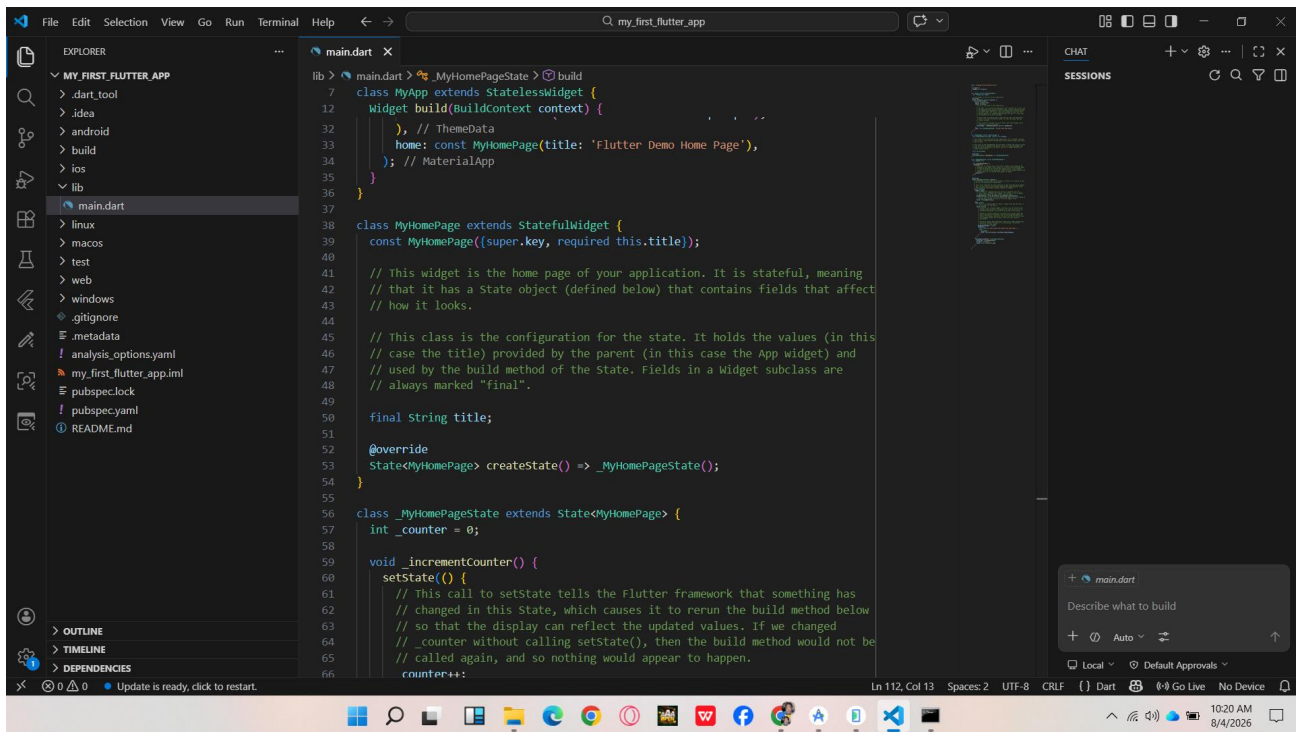


Flag 9 – Find the Entry Point

Figure out where the default Flutter page comes from.

Screenshot:

- The location of the file where the default page is from (the one showing on the emulator)



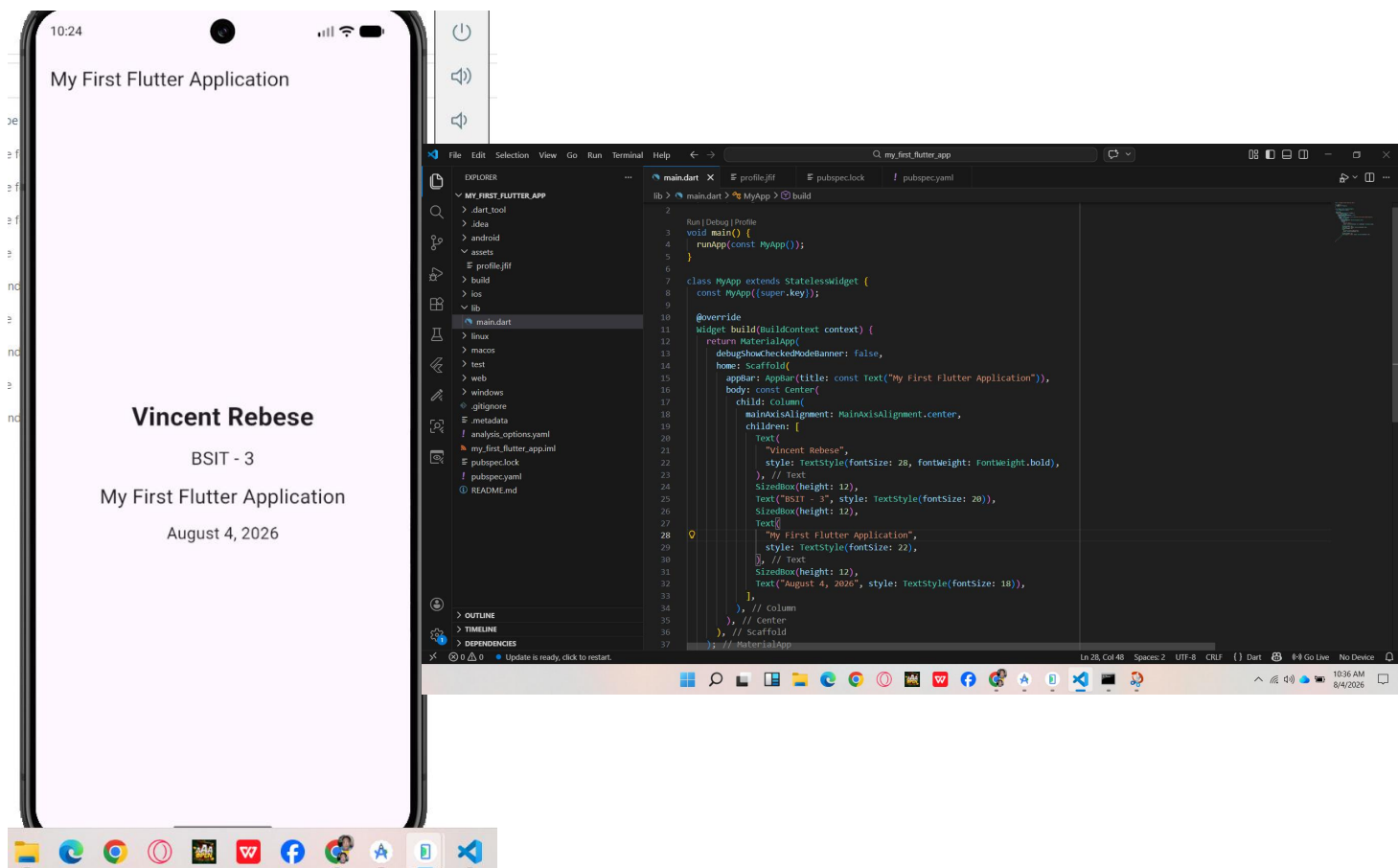
Flag 10 – Claim Your App

Replace the default screen with:

- Full Name
- Course & Section
- "My First Flutter Application"
- Today's Date

Screenshot:

- Your updated content of the code
- The updated UI



Flag 11 – Personalize It

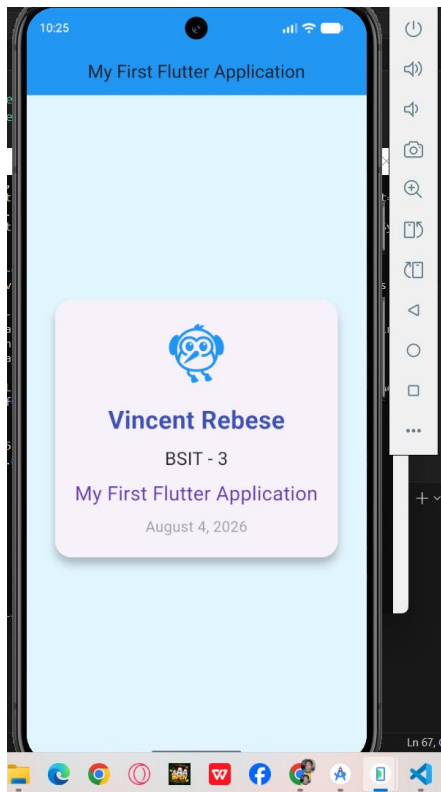
Research and implement at least **three** visual changes.

Examples:

- Change the background colour.
- Change the text colour.
- Centre the content.
- Increase font size.
- Add a border or rounded card.
- Change the font.
- Add an emoji or icon.

Screenshot:

- The code changed for personalizing
- The updated UI



Flag 12 – Add an Image

Research how to:

- Create an `assets` folder
- Register assets
- Display an image

Screenshot:

- The steps it took to add an image
- The updated UI

